Smart Motion-Sensing LED Bulb: How It Works

The Core Logic: How the Bulb Decides What to Do

The firmware runs a simple state machine. At any moment, the bulb is in one of these states:

Smart Idle: Bulb is off. The firmware reads the light sensor (LDR) to check if the room is dark. If it is dark and the motion sensor detects movement, the bulb turns on immediately and switches to Smart Active. If the room is bright, motion is completely ignored, you can wave your arms all day and nothing happens.

Smart Active: Bulb is on. The firmware stops reading the light sensor entirely. It does not matter if you turn on a room light or the sun comes up, the bulb stays on because it has already decided you needed light. The only thing that matters now is motion. Every time you move, a 60-second countdown resets. When you stop moving, and a 60-second window elapses without any movement, the bulb turns off and goes back to Smart Idle.

Manual Mode: You pressed Sensor Off on the remote. The light sensor and motion sensor are both disabled. The bulb becomes a simple remote-controlled light. Press Power on the remote to turn it on it stays on forever. Press Power again to turn it off. Nothing else affects it.

Forced On: You pressed Power once while in Smart Idle/Active (sensor enabled)
The bulb turns on and locks on. Motion is ignored, the light sensor is ignored, and the 60-second timer is paused.

How the Light Sensor Works the light sensor (LDR) is only read when the bulb is off. The moment the bulb turns on, the firmware stops reading it completely. This is critical, if the sensor is read while the bulb is on, you are measuring your own light, and you would think the room is bright, and that causes unnecessary instability

Pairing
How the Bulb and Remote Find Each Other

When you first power up the bulb, it enters pairing mode. The bulb LED blinks slowly**, one second on, two seconds off, for up to 60 seconds**. During this period, the device listens on a default/public radio channel for pairing requests, allowing any remote device to send a request that it will receive.
You press and hold the Power button on the remote for **3 seconds**. The remote wakes up, sends a pairing request on the public channel/default channel, and blinks its LED **one second ON, two seconds OFF** to indicate that it has entered a pairing mode and has sent a request while it waits for a response.

The bulb receives the request generates a random 32-bit token from the chip, stores that token along with the remote's address in permanent memory, and sends back a confirmation. The remote receives the confirmation, blinks **twice rapidly, and turns off** to show it worked, and switches to a private channel that only the paired bulb listens to.
While the bulb led **blinks (3 seconds on, 2 off, 3 on) Paired successfully**. Normal operation starts.

Paring is complete. From then on, only that remote can control the bulb.

Assuming nobody presses anything for 60 seconds during the first power on, the bulb stops blinking and defaults to a **Sensor enabled mode** and responds to environmental changes normally.
You can still send Paring and factory reset request only but on the public channel, since the private channel has not be created, once a remote has successfully paired with the bulb, It will only receive factory reset request only on the public channel (can be used to clear all data on the bulb and reconnect a new bulb if the formerly paired get lost), but you cannot **change sensor mode or turn on or off the light** until you pair a remote to the bulb properly.

To pair again later, for example if you get a new remote, you hold Power for 6 seconds. **This factory resets the bulb**: it clears the old pairing, re-enters pairing mode, and waits for a new remote. The bulb blinks 6 times to confirm the reset.

What You See: LED Feedback

The bulb LED is dark almost all the time. It only blinks in four situations:

- Slow blink (1 second on, 2 seconds off) Pairing mode. Waiting for a remote.
- Double long blink (3 seconds on, 2 off, 3 on) Paired successfully. Normal operation starts.
- One short flash — Sensors turned back on.
- Two short flashes — Sensors turned off.

The remote LED is different; it blinks after every button press. One blink means the bulb received the command. Three fast blinks mean it failed. This way, you always know the remote is working.

The Remote: How It Saves Battery

The remote does not run a loop. It sleeps between every button press, completely off, drawing less than a micro amp. When you press any button, it wakes in 4 microseconds, processes the press, sends the radio command, waits for a confirmation, blinks the LED, and goes back to sleep.

The Power button has three meanings: tap to toggle the bulb, hold for 3 seconds to pair, and hold for 6 seconds to factory reset. The other two buttons, Sensor On and Sensor Off, do one thing each: tap only.

What Happens If You Lose the Remote?

There is no button on the bulb. If the remote is lost or destroyed, the bulb stays paired to it forever. It will still work but be stuck on sensor-enabled mode. To connect a new remote, you long press the Power button and hold for 6 seconds to clear the bulb memory and the paring cycle begins again